

Generation Interconnection System Impact Study Report

For Queue Project Q19-1216

Septa 138 kV
March 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1216. The Interconnection Customer proposes a 185.0 MW natural gas combined cycle facility interconnecting at the Septa 138 kV substation in the Pennsylvania area. The request was received on May 28, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,799,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Nwbury - Septa 138.0 kV	Lighthouse Hill - Aes Beaver Vly Partners Beaver Vly 345.0 kV L	166.0	188	63.4
Havelock - Waukena 138.0 kV	Stewart Mountain Dam - Lighthouse Hill 345.0 kV	163.3	178	9.9
Arkoma - Waukena 138.0 kV line	Stewart Mountain Dam Substation - Lighthouse Hill 345.0 kV	149.2	203	9.9
Arkoma - Lighthouse Hill 138.0 kV	Lighthouse Hill - Stewart Mountain Dam 345.0 kV line	143.3	228	9.9
Blountsville - Havelock 138.0 kV	Lighthouse Hill - Stewart Mtn Dam 345.0 kV L	139.9	196	9.9
Hardee Power Station - Conrath Dpc Solar 138.0 kV	Conrath Dpc Solar - Sierra Brooks 138.0 kV	133.7	249	5.7
Walmart - Hardee Power Station 138.0 kV L	SierraBrooks - Conrath Dpc Solar 138.0 kV Line	133.5	242	5.7
Torrey - Toston 138.0 kV	Stewart Mtn Dam - Ritter Park 345.0 kV line	131.2	224	6.8
Kyles Ford - Pittsford 138.0 kV	Brunot Island - Conrath Dpc Solar 138.0 kV line	123.2	207	6.4
Bullard - Toston 138.0 kV	Stewart Mountain Dam - Ritter Park 345.0 kV	119.2	235	6.8
Eight Mile Crk - Newbury 138.0 kV	Lighthouse Hill - Aes Beaver Vly Partners Beaver Valley 345.0 kV	117.4	173	30.5
Torrey Substation - Blountsville 138.0 kV	Ritter Park - Stewart Mtn Dam 345.0 kV Line	116.0	264	6.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bullard - Ritter Park 138.0 kV	Stewart Mountain Dam - Ritter Park 345.0 kV	107.0	248	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nwbury - Septa 138.0 kV	Rebuild line with higher-capacity conductor	\$5,215,000	73% (\$3,812,000)
Havelock - Waukena 138.0 kV	Rebuild line with higher-capacity conductor	\$7,526,000	100% (\$7,526,000)
Arkoma - Waukena 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,814,000	100% (\$1,814,000)
Arkoma - Lighthouse Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$8,542,000	100% (\$8,542,000)
Blountsville - Havelock 138.0 kV	Rebuild line with higher-capacity conductor	\$2,683,000	100% (\$2,683,000)
Conrath Dpc Solar - Hardee Power Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$716,000	5% (\$37,000)
Hardee Power Sta - Walmart 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,099,000	5% (\$214,000)
Torrey - Toston 138.0 kV	Rebuild line with higher-capacity conductor	\$2,764,000	31% (\$845,000)
Kyles Ford - Pittsford 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,513,000	5% (\$117,000)
Bullard - Toston 138.0 kV	Rebuild line with higher-capacity conductor	\$2,864,000	31% (\$876,000)
Eight Mile Crk - Newbury 138.0 kV	Rebuild line with higher-capacity conductor	\$1,049,000	60% (\$629,000)
Blountsville - Torry 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,944,000	31% (\$594,000)
Bullard - Ritter Park 138.0 kV	Reconductor line	\$359,000	31% (\$110,000)
	Total allocated to project		\$27,799,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.